A positive proportion of primes are isolated

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Abstract

For all fixed $C_1, C_2 > 0$ we show that the set of indices n with $p_n - p_{n-1} > C_1 \log p_n$ and $p_{n+1} - p_n > C_2 \log p_n$ has positive lower density. Equivalently, for every fixed $C > 0$ a positive proportion of primes p are the only prime in $[p - C \log p,\, p + C \log p]$. As a consequence, the indices n for which $p_{n-1}/(n-1) < p_n/n < p_{n+1}/(n+1)$ have positive lower density, a density strengthening of the (already known) infinitude of such increasing triples, which is asked about with Erdős Problem #968. The method is entirely that of the recent preprint *Positive lower density of large prime gaps* (OpenAI, 2026), which proves the one-sided statement. Our contribution is the adaptation: a three-block geometry with a detection block placed between two suppressed blocks, and the use of the off-diagonal part of that paper's detector estimate to force *exactly one* prime in the detection block. We state precisely which results of the earlier paper are used as black boxes and give complete proofs of the modified statements.

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1 Introduction

Let p_n denote the nth prime. The prime number theorem identifies $\log p$ as the mean spacing of primes near p. A recent preprint [6], released by OpenAI and referred to below as *[O]*, proves that for every fixed $C > 0$ a positive proportion of consecutive gaps $p_{n+1} - p_n$ exceed $C \log p_n$, with ordinary lower density [6, Theorem 1.1]. It deduces that the indices n with $p_n/n < p_{n+1}/(n+1)$ have positive lower density, answering a question of Erdős and Prachar [2], [6, Corollary 1.2].

In this paper we show that the method of [O] controls two consecutive gaps simultaneously, on both sides of the same prime.

Theorem 1.1. *For all fixed real numbers $C_1, C_2 > 0$ there are $c = c(C_1, C_2) > 0$ and $N_0 = N_0(C_1, C_2)$ such that for every integer $N \geq N_0$,*

$$\#\big\{2 \leq n \leq N : \ p_n - p_{n-1} > C_1 \log p_n \ \ \textit{and} \ \ p_{n+1} - p_n > C_2 \log p_n\big\} \ \geq \ cN.$$

Call a prime p *C-isolated* if p is the only prime in the closed interval $[p - C\log p,\, p + C\log p]$. Taking $C_1 = C_2 = C$ in Theorem 1.1 gives:

Corollary 1.2. *For every fixed $C > 0$ there is $c(C) > 0$ such that, for all sufficiently large x, at least $c(C)\pi(x)$ primes $p \leq x$ are C-isolated.*

Proof. If $p = p_n$ with $p_n - p_{n-1} > C \log p_n$ and $p_{n+1} - p_n > C \log p_n$, then $p_{n-1} < p - C\log p$ and $p_{n+1} > p + C \log p$, so p is C-isolated. Apply Theorem 1.1 with $N = \pi(x)$. □

Theorem 1.1 contains the one-sided result [6, Theorem 1.1] (take C_1 arbitrary). The corresponding consequence for the ratios $u_n = p_n/n$ concerns two consecutive increases.

Corollary 1.3. *The set*

$$\left\{n \geq 2 : \ \frac{p_{n-1}}{n-1} < \frac{p_n}{n} < \frac{p_{n+1}}{n+1}\right\}$$

has positive lower asymptotic density. In particular, writing $u_n = p_n/n$, there are infinitely many n with $u_n < u_{n+1} < u_{n+2}$.

The question whether $u_n < u_{n+1} < u_{n+2}$ (or $u_n > u_{n+1} > u_{n+2}$) holds for infinitely many n is recorded on the `erdosproblems.com` page of Problem #968 [3] (consulted October 2026), together with the Erdős–Prachar question on the density of $\{n : u_n < u_{n+1}\}$, as a further question of Erdős; see [2] for the original context. We have checked the statement of that question as displayed on the page but not its exact original source, and we make no claim beyond the displayed formulation. The displayed (infinitude) question is in fact already settled in both directions by earlier work: infinitely many pairs of consecutive gaps both much larger than $\log p_n$ follow from results of Erdős, Maier and Ford–Maynard–Tao on chains of large gaps, and the decreasing case follows from Maynard's bounded gaps theorem (see Remark 2.4). The new content of Corollary 1.3 is the positive lower density of increasing triples, which does not follow from those results. We have not checked whether a positive density of decreasing triples follows from known positive-proportion results for small gaps (Goldston–Pintz–Yıldırım, Maynard); this seems likely.

What is new, and what is taken from [O]

All analytic ingredients come from [O]: the square Z^2 of a signed sum of smooth divisor sums attached to a set of shifts, the uniform mixed-moment formula [6, Proposition 5.1], the box average of the singular series [6, Lemma 5.4], the prime-deletion identity [6, (8)], and the cancellation between adjacent dimensions [6, Proposition 4.2]. We do not reprove these. The new points are the following.

(i) *Geometry.* In [O] the detection block J lies to the left of the suppressed block I, and the last prime of J starts a long gap. Here the shift set is a union $I = I_L \cup I_R$ of two blocks with the detection block J between them. If $m + J$ contains *exactly one* prime and $m + I$ contains none, that prime is isolated on both sides.

(ii) *Exactly one prime.* The weighted probability of the event "exactly one prime in $m + J$" is bounded below via $\mathbf{1}_{\{N=1\}} \geq N - N(N-1)$. The negative term is controlled by the *off-diagonal* part of the detector estimate of [O], whose limit is $c_G^2 \eta^2 w$ with a constant c_G depending on neither the weight family nor the length of I. This is quadratic in the relative length η of J, while the main term is linear, so a short block J suffices.

(iii) *Moment identities for the new configuration.* We restate the moment identities [6, Proposition 3.1] for a non-contiguous shift set and a detection block of different length, and prove them (Proposition 4.1) from [6, Proposition 5.1] and a block-decomposed form of [6, Lemma 5.4] (Lemma 3.4).

The constants obtained are not explicit: as in [O], the second moment $\mathbb{E}_X W^2$ is controlled only for each fixed weight family, and the family depends on C_1, C_2. In Section 6 we explain, as remarks only, why the method does not seem to reach three consecutive large gaps, or a large gap next to a small one.

Organization

Section 2 fixes the geometry and deduces Theorem 1.1 and Corollary 1.3 from a weight proposition (Proposition 2.1). Section 3 recalls the construction and the black-box results of [O]. Section 4 proves the moment identities for the three-block configuration, and Section 5 proves Proposition 2.1. Section 6 contains remarks on limitations.

Notation

X tends to infinity through integers, $L = \log X$, and $\mathbb{E}_X A = X^{-1} \sum_{X < m \leq 2X} A(m)$. We write $\vartheta(n) = (\log n)\mathbf{1}_{\{n \text{ prime}\}}$, μ is the Möbius function, and $\|\cdot\|_2$ is the L^2 norm on a nonnegative orthant (absolute value in dimension zero). All parameters other than X are fixed before $X \to \infty$; implied constants and rates of convergence may depend on them.

2 The three-block geometry and the deduction of the theorem

2.1 Blocks

Fix a real number $\eta > 0$ and an integer $R \geq 1$. Put

$$h = \lfloor \eta L \rfloor, \qquad B_i = \{(i-1)h + 1, \dots, ih\} \quad (1 \leq i \leq 2R+1),$$

and

$$I_L = B_1 \cup \cdots \cup B_R, \qquad J = B_{R+1}, \qquad I_R = B_{R+2} \cup \cdots \cup B_{2R+1}, \qquad I = I_L \cup I_R.$$

Thus $I_L = \{1, \dots, Rh\}$, $J = \{Rh + 1, \dots, (R+1)h\}$ and $I_R = \{(R+1)h + 1, \dots, (2R+1)h\}$; the sets I and J are disjoint, $|J| = h$, $|I| = 2Rh$, and every shift lies in $[1, (2R+1)h]$. We write

$$\lambda = 2R\eta, \qquad \text{so that} \qquad \frac{|I|}{L} \to \lambda, \qquad \frac{|J|}{L} \to \eta.$$

For $B \in \{I, J\}$ and $X < m \le 2X$ define

$$N_B(m) = \#\{b \in B : \ m + b \text{ prime}\}, \qquad V_B(m) = \frac{1}{L} \sum_{b \in B} \vartheta(m + b).$$

Since every argument $m + b$ exceeds X, we have $\vartheta(m + b) \ge L$ when $m + b$ is prime, hence

$$N_B(m) \le V_B(m) \le q_X N_B(m), \qquad q_X := \frac{\log(2X + (2R + 1)h)}{L} = 1 + o(1). \tag{1}$$

2.2 The majorant for pairs

Put $\tau = 1/8$, as in [O]. Fix once and for all a real function $G \in C_c^\infty(\mathbb{R})$ with $G(0) = 1$ and $G(u) = 0$ for $u \ge \tau$, and set

$$c_G = \int_0^\infty G'(u)^2 \, du.$$

This is the function and the constant of [6, Proposition 3.1]. Since $\int_0^\tau G' = -1$, Cauchy–Schwarz gives $c_G \ge 1/\tau = 8$; in particular $c_G > 0$. For a shift b write

$$D_G(m; b) = \sum_{d \mid m+b} \mu(d) \, G\Big(\frac{\log d}{L}\Big).$$

If $m + b$ is a prime exceeding X, the only divisors are 1 and $m + b$, and $\log(m + b)/L > 1 > \tau$, so $D_G(m; b) = G(0) = 1$. Consequently

$$\mathbf{1}_{\{m+b \text{ prime}\}} \le D_G(m; b)^2 \qquad (X < m \le 2X, \ b \ge 1). \tag{2}$$

Define

$$\Pi_J(m) = \sum_{\substack{b,c \in J \\ b \ne c}} D_G(m; b)^2 D_G(m; c)^2.$$

Since $N_J(N_J - 1) = \sum_{b \ne c \in J} \mathbf{1}_{\{m+b \text{ prime}\}} \mathbf{1}_{\{m+c \text{ prime}\}}$, (2) gives

$$N_J(m)\big(N_J(m) - 1\big) \le \Pi_J(m). \tag{3}$$

2.3 The weight proposition

Proposition 2.1 (Weights for three blocks)**.** *Fix $\eta > 0$ and an integer $R \ge 1$, and define the blocks as above. For every $\varepsilon > 0$ there are weights $W_X(m) \ge 0$, defined for integers $X < m \le 2X$, such that*

$$\mathbb{E}_X W_X \longrightarrow 1, \qquad \mathbb{E}_X(W_X V_J) \longrightarrow \eta, \tag{4}$$

$$\limsup_{X \to \infty} \mathbb{E}_X(W_X V_I) \le \varepsilon, \qquad \mathbb{E}_X(W_X \Pi_J) \longrightarrow c_G^2 \eta^2, \tag{5}$$

$$\mathbb{E}_X W_X^2 = O_{\eta,R,\varepsilon}(1). \tag{6}$$

The constant c_G depends only on the fixed function G; it does not depend on η, R or ε.

The proof is given in Section 5. The decisive feature is the last limit in (5): the weighted expected number of ordered pairs of distinct primes in $m + J$ is at most $c_G^2 \eta^2$, quadratic in η, whereas the weighted expected number of primes is η.

2.4 From starts to isolated primes

Lemma 2.2. *Let X be large, and let $\mathcal{E}_X$ be the set of integers $m \in (X, 2X]$ such that $m+J$ contains exactly one prime and $m+I$ contains no prime. For $m \in \mathcal{E}_X$ let $p(m)$ be the prime in $m+J$ and let $p(m)^-$, $p(m)^+$ be its consecutive prime neighbours. Then*

$$p(m) - p(m)^- > Rh, \qquad p(m)^+ - p(m) > Rh,$$

and at least $|\mathcal{E}_X|/h$ distinct primes in $(X,\, 2X + (R+1)h]$ are of the form $p(m)$ with $m \in \mathcal{E}_X$.

Proof. Write $p = p(m)$, so $m + Rh + 1 \le p \le m + (R+1)h$. There is no prime in $m + I_L = [m+1, m+Rh]$ and none in $m+J$ other than p. Hence $p^- \le m$, and $p - p^- \ge Rh + 1$. There is no prime in $m + I_R = [m + (R+1)h + 1,\, m + (2R+1)h]$ and none in $(p, m + (R+1)h]$, so $p^+ \ge m + (2R+1)h + 1$ and $p^+ - p \ge (2R+1)h + 1 - (R+1)h = Rh + 1$.

If $p(m) = p$ then $p - (R+1)h \le m \le p - Rh - 1$, an interval containing exactly h integers. Hence each prime equals $p(m)$ for at most h values of $m \in \mathcal{E}_X$, and the number of distinct values is at least $|\mathcal{E}_X|/h$. Finally $X < m < p \le 2X + (R+1)h$. □

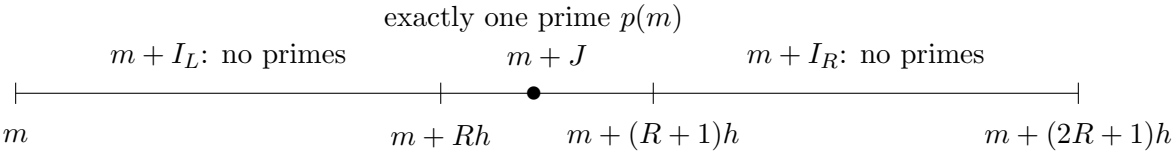


Figure 1: The three-block configuration. The unique prime in $m+J$ is at distance more than Rh from every other prime.

2.5 Proof of Theorem 1.1

Fix $C_1, C_2 > 0$ and put $C = \max\{C_1, C_2\}$. Make the following choices, in this order.

(a) $\tau = 1/8$ and G are fixed as above, hence so is $c_G \ge 8$.

(b) $\eta = 1/(2c_G^2)$. Then $\eta - c_G^2\eta^2 = \eta/2$.

(c) $R = \lfloor C/\eta \rfloor + 1$, so that $R\eta > C$.

(d) $\varepsilon = \eta/4$, and W_X is the weight family given by Proposition 2.1 for these η, R, ε.

Let $M < \infty$ satisfy $\mathbb{E}_X W_X^2 \le M$ for all large X, by (6).

Step 1: a pointwise inequality. For a nonnegative integer N we have $N - N(N-1) = N(2-N)$, which equals $0, 1, 0$ for $N = 0, 1, 2$ and is negative for $N \ge 3$. Hence $N - N(N-1) \le \mathbf{1}_{\{N=1\}}$. Let $\mathbf{1}_{\mathcal{E}}(m)$ be the indicator of $\{N_J(m) = 1,\ N_I(m) = 0\}$. We claim

$$\mathbf{1}_{\mathcal{E}} \;\ge\; N_J - N_J(N_J - 1) - N_I. \tag{7}$$

If $N_I = 0$, the right side is at most $\mathbf{1}_{\{N_J=1\}} = \mathbf{1}_{\mathcal{E}}$. If $N_I \ge 1$, it is at most $1 - 1 = 0 \le \mathbf{1}_{\mathcal{E}}$.

Step 2: weighted mass of the event. Multiply (7) by $W_X \ge 0$ and use (1) and (3):

$$W_X \mathbf{1}_{\mathcal{E}} \;\ge\; q_X^{-1} W_X V_J - W_X \Pi_J - W_X V_I.$$

Averaging and using (4), (5) and $q_X \to 1$,

$$\liminf_{X\to\infty} \mathbb{E}_X(W_X \mathbf{1}_{\mathcal{E}}) \;\geq\; \eta - c_G^2\eta^2 - \varepsilon = \frac{\eta}{2} - \frac{\eta}{4} = \frac{\eta}{4}.$$

Thus $\mathbb{E}_X(W_X\mathbf{1}_{\mathcal{E}}) \geq \eta/8$ for all large X.

Step 3: ordinary proportion. By Cauchy–Schwarz,

$$\frac{\eta}{8} \leq \mathbb{E}_X(W_X\mathbf{1}_{\mathcal{E}}) \leq (\mathbb{E}_X W_X^2)^{1/2}\Big(\frac{|\mathcal{E}_X|}{X}\Big)^{1/2}, \qquad \text{so} \qquad |\mathcal{E}_X| \geq \delta X, \quad \delta := \frac{\eta^2}{64M} > 0,$$

for all large X, where $\mathcal{E}_X$ is as in Lemma 2.2.

Step 4: isolated primes in a dyadic range. By Lemma 2.2 there are at least $\delta X/h \geq \delta X/(\eta L)$ distinct primes $p \in (X, 2X + (R+1)h]$ with $p - p^- > Rh$ and $p^+ - p > Rh$. For such p, $\log p \leq \log(3X) = L + O(1)$, while $Rh \geq R\eta L - R$. Since $R\eta > C$, for all large X,

$$Rh > C\log p \;\geq\; \max\{C_1, C_2\}\log p.$$

Writing $p = p_n$, we get $p_n - p_{n-1} > C_1 \log p_n$ and $p_{n+1} - p_n > C_2 \log p_n$.

Step 5: initial segments. Let N be large and put $X = \lfloor p_N/3 \rfloor$. Then $2X + (R+1)h \leq 2p_N/3 + O(\log p_N) < p_N$, so every prime found in Step 4 is p_n with $\pi(X) < n < N$; in particular $n \geq 2$. By the prime number theorem $X/\log X \sim p_N/(3\log p_N) \sim N/3$, so $X/L \geq N/4$ for large N. Hence the count in Theorem 1.1 is at least $\delta X/(\eta L) \geq \delta N/(4\eta)$. Thus $c = \delta/(4\eta)$ is admissible, and N_0 is determined by the finitely many "sufficiently large" conditions above, all of which depend only on C_1, C_2 through the choices (a)–(d). □

Remark 2.3. Compared with [6, Section 2], the only change in the counting is that the event "$m + J$ contains a prime" is replaced by "$m + J$ contains exactly one prime". In [O] the former is detected by Cauchy–Schwarz, $\mathbb{E}_X(W\mathbf{1}_{V_J>0}) \geq (\mathbb{E}_X WV_J)^2/\mathbb{E}_X(WV_J^2)$, which gives no information about the number of primes in $m + J$. If $m + J$ contained two primes, the first would have a long gap only to its left and the last only to its right. The inclusion–exclusion bound (7) needs the second factorial moment of N_J to be $o(\eta)$ as $\eta \to 0$, which is exactly what the off-diagonal detector term supplies.

2.6 Proof of Corollary 1.3

Write $u_n = p_n/n$ and $d_n = p_{n+1} - p_n$. For $n \geq 1$,

$$u_n < u_{n+1} \iff np_{n+1} > (n+1)p_n \iff n(p_{n+1} - p_n) > p_n \iff d_n > \frac{p_n}{n}.$$

Applying this with $n - 1$ in place of n (for $n \geq 2$),

$$u_{n-1} < u_n \iff d_{n-1} = p_n - p_{n-1} > \frac{p_{n-1}}{n-1}.$$

By the prime number theorem, $p_k/k \sim \log p_k$ as $k \to \infty$. Hence there is n_1 such that for all $n \geq n_1$,

$$\frac{p_n}{n} < 2\log p_n, \qquad \frac{p_{n-1}}{n-1} < 2\log p_{n-1} < 2\log p_n.$$

Let $n \geq n_1$ belong to the set counted in Theorem 1.1 with $C_1 = C_2 = 2$. Then $d_{n-1} > 2\log p_n > p_{n-1}/(n-1)$ and $d_n > 2\log p_n > p_n/n$, so $u_{n-1} < u_n < u_{n+1}$. Therefore, for large N,

$$\#\{2 \leq n \leq N :\; u_{n-1} < u_n < u_{n+1}\} \;\geq\; c(2,2)N - n_1 \;\geq\; \tfrac{1}{2}c(2,2)N,$$

which is positive lower density. Replacing n by $n + 1$ gives infinitely many n with $u_n < u_{n+1} < u_{n+2}$. □

Remark 2.4. The decreasing pattern $u_n > u_{n+1} > u_{n+2}$ is equivalent to $d_n < p_n/n$ and $d_{n+1} < p_{n+1}/(n+1)$. Since $p_k/k \to \infty$, it holds for all large n with $p_{n+2} - p_n$ bounded. Maynard's theorem [5] gives $\liminf_n (p_{n+2} - p_n) < \infty$, so the decreasing pattern also occurs infinitely often. This is a simple observation and we make no claim about positive density in the decreasing case; it is not reached by the method of this paper (see Section 6).

3 The construction of [O] and the results used as black boxes

In this section we recall the definitions of [O] that we need and state the three results used without proof. The only change in the definitions is that the shift set I is now the non-contiguous set $I_L \cup I_R$ of Section 2, of size $|I| = 2Rh$.

3.1 Divisor sums and the square

For a compactly supported $F : [0,\infty)^j \to \mathbb{R}$ and a tuple $\mathbf{b} = (b_1, \dots, b_j)$ of distinct shifts,

$$D_F(m; \mathbf{b}) = \sum_{d_i \mid m + b_i \ (1 \le i \le j)} \mu(d_1) \cdots \mu(d_j)\, F\Big(\frac{\log d_1}{L}, \dots, \frac{\log d_j}{L}\Big),$$

with $D_F = F$ a scalar when $j = 0$. If F is symmetric we write $D_F(m; S)$ for an unordered set S of j shifts.

Let $k \ge 1$ be an integer. For $1 \le j \le k$ let

$$f_j \in C_c^\infty\big(\{\mathbf{t} \in (0,\infty)^j : \ t_1 + \cdots + t_j < \tau\}\big)$$

be real and symmetric, let $f_0 \in \mathbb{R}$, and put $f_{k+1} = 0$. Define

$$F_j(\mathbf{v}) = \int_{s_i \ge v_i \ (1 \le i \le j)} f_j(\mathbf{s})\, d\mathbf{s}, \qquad F_0 = f_0, \qquad F_{k+1} = 0, \qquad (T f_{j+1})(\mathbf{t}) = \int_0^\infty f_{j+1}(\mathbf{t}, u)\, du.$$

Then F_j is symmetric, vanishes on $[0,\infty)^j$ outside $\{\sum v_i < \tau\}$, admits a smooth compactly supported extension to $\mathbb{R}^j$ (see the beginning of the proof of [6, Proposition 3.1]), and

$$(-1)^j \partial_1 \cdots \partial_j F_j = f_j \qquad \text{on } [0,\infty)^j, \tag{8}$$

which is [6, (4)]. As f_j has compact support in the open orthant, the right side vanishes whenever some coordinate is 0 (for $j \ge 1$). We call a family $\mathbf{f} = (f_j)_{0 \le j \le k}$ with these properties an *admissible family*. Define

$$Z(m) = \sum_{j=0}^{k} \sum_{\substack{S \subset I \\ |S| = j}} D_{F_j}(m; S), \qquad \alpha_j = \frac{\lambda^j}{j!} \quad (\lambda = 2R\eta),$$

and the two quadratic forms of [O],

$$w = w(\mathbf{f}) = \sum_{j=0}^{k} \alpha_j \|f_j\|_2^2, \qquad v = v(\mathbf{f}) = \lambda \sum_{j=0}^{k} \alpha_j \|f_j + T f_{j+1}\|_2^2. \tag{9}$$

In dimension zero, $\|f_0 + T f_1\|_2^2 = |f_0 + \int_0^\infty f_1|^2$.

3.2 Prime deletion

The following identity is [6, (8)]; we recall its short proof since it is the only place where the arithmetic of the shift set I enters, and it is insensitive to the shape of I.

Lemma 3.1 (Prime deletion)**.** *Let $a \in I$ and put $\widetilde{F}_j(\mathbf{v}) = F_j(\mathbf{v}) + F_{j+1}(\mathbf{v}, 0)$ for $0 \le j \le k$. If $X < m \le 2X$ and $m + a$ is prime, then*

$$Z(m) = Z_a(m) := \sum_{j=0}^{k} \sum_{\substack{U \subset I \setminus \{a\} \\ |U| = j}} D_{\widetilde{F}_j}(m; U).$$

Moreover $\widetilde{F}_j$ is symmetric, vanishes on the orthant outside $\{\sum v_i < \tau\}$, has a smooth compactly supported extension, and $(-1)^j \partial_1 \cdots \partial_j \widetilde{F}_j = f_j + T f_{j+1}$; for $j \ge 1$ this derivative vanishes whenever a coordinate is 0.

Proof. Let $S \ni a$ with $|S| = j + 1$. In $D_{F_{j+1}}(m; S)$ the divisor d_a of the prime $m + a$ is 1 or $m + a$. In the second case $\log d_a / L > 1 > \tau$, so F_{j+1} vanishes. Hence $D_{F_{j+1}}(m; S) = D_{F_{j+1}(\cdot, 0)}(m; S \setminus \{a\})$, using symmetry to place the a-coordinate last. Collecting, for each $U \subset I \setminus \{a\}$, the term from $S = U$ and the term from $S = U \cup \{a\}$ gives the identity. Next, $F_{j+1}(\mathbf{v}, 0) = \int_{\mathbf{s} \ge \mathbf{v}} (T f_{j+1})(\mathbf{s})\, d\mathbf{s}$, so its signed mixed derivative is $T f_{j+1}$; the remaining properties follow as for F_j, since $T f_{j+1}$ is smooth with compact support in the open orthant intersected with $\{\sum t_i < \tau\}$. □

3.3 The mixed-moment formula

For a finite set $\mathcal{H}$ of distinct integers put $\nu_p(\mathcal{H}) = |\mathcal{H} \bmod p|$ and

$$\mathfrak{S}(\mathcal{H}) = \prod_p \Big(1 - \frac{\nu_p(\mathcal{H})}{p}\Big)\Big(1 - \frac{1}{p}\Big)^{-|\mathcal{H}|}, \qquad \mathfrak{S}(\varnothing) = 1.$$

This is nonnegative, and $\mathfrak{S}(\mathcal{H}) = 1$ when $|\mathcal{H}| = 1$.

Proposition 3.2 ([6, Proposition 5.1])**.** *Fix nonnegative integers $j_1, \dots, j_{R'}$ and real functions F_r on $[0, \infty)^{j_r}$, each with a smooth compactly supported extension to $\mathbb{R}^{j_r}$ and vanishing on the orthant outside $\{v_1 + \cdots + v_{j_r} \le \rho_r\}$, with fixed $\rho_r \ge 0$ (scalars when $j_r = 0$). Let $\Gamma = \{(r, \ell) : 1 \le \ell \le j_r\}$ and fix a surjection $\kappa : \Gamma \to \{1, \dots, t\}$ with fibres $\mathcal{M}_i = \kappa^{-1}(i)$. Let $a_1, \dots, a_t$ be distinct integers with $|a_i| \le DL^B$ ($D > 0$, $B \ge 1$ fixed), and set $b_{r,\ell} = a_{\kappa(r,\ell)}$. Let $\delta \in \{0, 1\}$; if $\delta = 1$ let a_0 be one further integer, distinct from $a_1, \dots, a_t$, with $|a_0| \le DL^B$. Put $\chi_0 = 1$, $\mathcal{H} = \{a_1, \dots, a_t\}$ if $\delta = 0$, and $\chi_1(m) = \vartheta(m + a_0)$, $\mathcal{H} = \{a_0, a_1, \dots, a_t\}$ if $\delta = 1$. Assume $\sum_r \rho_r < 1$ if $\delta = 0$ and $\sum_r \rho_r < 1/2$ if $\delta = 1$. Then there is a real constant $\mathcal{C}$, depending only on the functions and on κ, such that*

$$\mathbb{E}_X \Big[\chi_\delta(m) \prod_r D_{F_r}(m; \mathbf{b}^{(r)})\Big] = L^{-t}\Big\{\mathfrak{S}(\mathcal{H})\, \mathcal{C} + O\big(L^{-1/2} (\log L)^A\big)\Big\}, \tag{10}$$

with A depending only on $\sum_r j_r$, and the implied constant depending on the fixed data but uniform in the shifts. If $|\mathcal{M}_i| \le 2$ for all i, then

$$\mathcal{C} = \int \prod_r \Big((-1)^{j_r} \partial_1 \cdots \partial_{j_r} F_r\Big)\big((y_{\kappa(r,\ell)})_{\ell=1}^{j_r}\big) \prod_{i:\, |\mathcal{M}_i| = 2} dy_i, \tag{11}$$

where $y_i = 0$ when $|\mathcal{M}_i| = 1$ and the integrated variables range over $[0, \infty)$. When $\Gamma = \varnothing$, $t = 0$ and $\mathcal{C} = \prod_r F_r$.

(We write R' for the number of factors to avoid a clash with our R.) The proof in [O] uses the Bombieri–Vinogradov theorem in the form of [1] and is valid for arbitrary shifts satisfying the stated bound; in particular it is indifferent to whether the shifts come from contiguous blocks. In our application all shifts lie in $[1,(2R+1)h] \subset [1,(2R+1)\eta L]$, so we may take $B=1$ and $D=(2R+1)\eta+1$.

3.4 The singular series in boxes

Lemma 3.3 ([6, Lemma 5.4]; a box form of Gallagher's mean [4])**.** *Fix an integer $s \geq 0$ and $K \geq 1$. For each positive integer h let $I_1,\dots,I_s$ be intervals of h consecutive integers, all contained in an interval of diameter at most Kh. Then, as $h\to\infty$,*

$$\sum_{\substack{a_i\in I_i\ (1\leq i\leq s)\\ a_1,\dots,a_s\ \text{distinct}}} \mathfrak{S}(\{a_1,\dots,a_s\}) = h^s(1+o(1)),$$

uniformly in the intervals subject to these conditions.

The intervals I_i in Lemma 3.3 need not be distinct. We use it in the following block-decomposed form, which is the only new input on the singular series.

Lemma 3.4 (Unions of blocks)**.** *Fix $s\geq 0$ and, for $1\leq i\leq s$, a nonempty set $\mathcal{U}_i \subset \{1,\dots,2R+1\}$ of block indices, and put $U_i=\bigcup_{\ell\in\mathcal{U}_i} B_\ell$. Then, as $X\to\infty$,*

$$\sum_{\substack{a_i\in U_i\ (1\leq i\leq s)\\ a_1,\dots,a_s\ \text{distinct}}} \mathfrak{S}(\{a_1,\dots,a_s\}) = |U_1|\cdots|U_s|\,(1+o(1)).$$

In particular this holds when each U_i is I or J.

Proof. The blocks $B_1,\dots,B_{2R+1}$ are pairwise disjoint, so every tuple $(a_1,\dots,a_s)\in U_1\times\cdots\times U_s$ determines a unique $(\ell_1,\dots,\ell_s)\in\mathcal{U}_1\times\cdots\times\mathcal{U}_s$ with $a_i\in B_{\ell_i}$. Distinctness is a property of the tuple. Hence the sum equals

$$\sum_{(\ell_i)\in\prod_i\mathcal{U}_i}\ \sum_{\substack{a_i\in B_{\ell_i}\\ \text{distinct}}} \mathfrak{S}(\{a_1,\dots,a_s\}).$$

Each B_ℓ consists of h consecutive integers, and all lie in $[1,(2R+1)h]$, of diameter less than $(2R+1)h$. Lemma 3.3 with $K=2R+1$ shows that each inner sum is $h^s(1+o(1))$ uniformly as $h=\lfloor\eta L\rfloor\to\infty$. There are $\prod_i|\mathcal{U}_i|$ outer terms, a fixed number, and $\prod_i|\mathcal{U}_i|\,h^s=\prod_i|U_i|$. □

3.5 Cancellation between adjacent dimensions

Proposition 3.5 ([6, Proposition 4.2])**.** *For every fixed $\lambda>0$ there are admissible families $\mathbf{f}$, with increasing finite maximum dimension k, for which $w(\mathbf{f})\to 1$ and $v(\mathbf{f})\to 0$, where w,v are given by (9).*

This is a statement about the constants w and v only, before any limit in X; it involves the shift configuration solely through the number λ, which in our application is $\lambda=2R\eta$. The families constructed in [O] are of the form $f_j=(-1)^j(r\alpha_j)^{-1/2}Q_j\chi(S_j)$ for $k-r<j\leq k$ and $f_j=0$ otherwise, with Q_j a product of a fixed nonnegative profile; they are real, symmetric, smooth, and supported in compact subsets of $\{\mathbf{t}\in(0,\infty)^j:\sum t_i<\tau\}$, hence admissible in the above sense.

4 Moment identities for the three-block configuration

The following proposition is the analogue of [6, Proposition 3.1] for our configuration. The differences are: the shift set I is a union of $2R$ blocks; the detection block J lies between two of them and has a length ηL unrelated to $|I| \approx \lambda L$; and instead of the full detector bound [6, (7)] we record the exact limit of its off-diagonal majorant.

Proposition 4.1. *Fix $\eta > 0$, $R \geq 1$, $k \geq 1$ and an admissible family $\mathbf{f}$. Then, as $X \to \infty$,*

$$\mathbb{E}_X Z^2 \longrightarrow w, \qquad \mathbb{E}_X Z^4 = O(1), \tag{12}$$

$$\mathbb{E}_X(Z^2 V_J) \longrightarrow \eta w, \qquad \mathbb{E}_X(Z^2 V_I) \longrightarrow v, \tag{13}$$

$$\mathbb{E}_X(Z^2 \Pi_J) \longrightarrow c_G^2 \eta^2 w, \tag{14}$$

with w, v as in (9) (with $\lambda = 2R\eta$) and c_G as in Section 2.

4.1 Reduction to equality patterns

All five quantities are finite linear combinations of averages

$$\mathbb{E}_X\Big[\chi_\delta(m) \prod_r D_{F_r}(m; \mathbf{b}^{(r)})\Big] \tag{15}$$

of the type in Proposition 3.2, summed over shifts. We organize such sums as follows.

Each factor D_{F_r} is $D_{F_j}(m;S)$, $D_{\widetilde{F}_j}(m;U)$ or $D_G(m;b)$; the underlying functions have support budget $\tau = 1/8$ (G vanishes on $[\tau,\infty)$). Suppose the factor D_{F_r} runs over j_r-element subsets of a set $U^{(r)} \in \{I, J\}$ (with $j_r = 1$ and $U^{(r)} = J$ for D_G). For a given tuple of subsets, the coincidences between their elements determine a surjection κ as in Proposition 3.2, the *equality pattern*. Since the functions are symmetric, the sum over subsets with a given pattern equals $1/\mathfrak{n}_\kappa$ times the sum over tuples $(a_1, \dots, a_t)$ of *distinct* integers with a_i in a prescribed set $U_i \in \{I, J\}$, where $\mathfrak{n}_\kappa$ is the number of tuples representing the same configuration of subsets (a product of factorials depending only on κ). There are finitely many patterns, since all dimensions are at most k and the number of factors is at most six.

Lemma 4.2 (Summing one pattern)**.** *Fix a pattern κ with t distinct unmarked shifts and prescribed sets $U_1, \dots, U_t \in \{I, J\}$, and, in the marked case, a prescribed set $U_0 \in \{I, J\}$ for a_0. Assume the support condition of Proposition 3.2. Let Σ_κ denote the sum of (15) over distinct $(a_1, \dots, a_t)$ with $a_i \in U_i$ (and over $a_0 \in U_0$ distinct from them, in the marked case), multiplied by $L^{-\delta}$. Then*

$$\Sigma_\kappa = \mathcal{C}_\kappa \, \frac{|U_0|^\delta \, |U_1| \cdots |U_t|}{L^{t+\delta}} + o(1),$$

where $\mathcal{C}_\kappa$ is the constant of Proposition 3.2. In particular $\Sigma_\kappa = O(1)$, and $\Sigma_\kappa = o(1)$ if $\mathcal{C}_\kappa = 0$.

Proof. By (10), $\Sigma_\kappa = L^{-t-\delta}\mathcal{C}_\kappa \sum \mathfrak{S}(\mathcal{H}) + E$, where the sum is over the admissible shift tuples and $|E| \ll L^{-t-\delta} \cdot \#\{\text{tuples}\} \cdot L^{-1/2}(\log L)^A$. The number of tuples is at most $((2R+1)h)^{t+\delta} = O(L^{t+\delta})$, so $E = O(L^{-1/2}(\log L)^A) = o(1)$. By Lemma 3.4 (applied with $s = t + \delta$, the entries of $\mathcal{H}$ being $a_0, a_1, \dots, a_t$), $\sum \mathfrak{S}(\mathcal{H}) = |U_0|^\delta |U_1| \cdots |U_t|(1 + o(1))$, and this is $O(L^{t+\delta})$. □

This is the summation principle stated after [6, Lemma 5.4], with Lemma 3.4 in place of [6, Lemma 5.4].

Lemma 4.3 (Matching)**.** *Consider a product of two factors $D_\Phi(m;S)\,D_\Psi(m;T)$, where $S,T\subset I$, $|S|=j$, $|T|=j'$, and Φ,Ψ are symmetric with signed mixed derivatives ϕ,ψ that vanish whenever a coordinate is 0 (when $j\geq 1$, resp. $j'\geq 1$), possibly multiplied by further factors whose shifts lie in J and whose fibres all have size two. If the resulting pattern has all fibres of size at most two and $S\neq T$, then $\mathcal{C}_\kappa=0$. If $S=T$, the coordinates of S and T are paired and $\mathcal{C}_\kappa=\int_{[0,\infty)^j}\phi\psi\,d\mathbf{y}$ times the contribution of the further factors.*

Proof. If $S\neq T$, some shift lies in exactly one of S,T and in no further factor (the further factors have shifts in J, disjoint from I). Its fibre is a singleton, so in (11) the corresponding coordinate of ϕ or ψ is set to 0, and the integrand vanishes identically. If $S=T$, every shift of S forms a fibre of size two, and (11) factors as stated by Fubini (the further factors involve disjoint variables). □

4.2 Proof of Proposition 4.1

Unmarked second moment. Expanding, $\mathbb{E}_X Z^2=\sum_{j,j'}\sum_{S,T}\mathbb{E}_X[D_{F_j}(m;S)D_{F_{j'}}(m;T)]$ with $S,T\subset I$. The two factors have budgets summing to $2\tau<1$, and each fibre has size one or two. By Lemmas 4.2 and 4.3 with (8), all patterns with $S\neq T$ contribute $o(1)$. For $S=T$, $|S|=j$, the pattern has $t=j$ shifts, $\mathfrak{n}_\kappa=j!$, all $U_i=I$, and $\mathcal{C}_\kappa=\|f_j\|_2^2$ (for $j=0$ the average is exactly f_0^2). Hence

$$\mathbb{E}_X Z^2=\sum_{j=0}^{k}\frac{\|f_j\|_2^2}{j!}\frac{|I|^j}{L^j}+o(1)\longrightarrow\sum_{j=0}^{k}\frac{\lambda^j}{j!}\|f_j\|_2^2=w.$$

Marked moment in J. We have $\mathbb{E}_X(Z^2V_J)=L^{-1}\sum_{a_0\in J}\mathbb{E}_X[\vartheta(m+a_0)Z(m)^2]$. Here $\delta=1$, the budgets sum to $2\tau=1/4<1/2$, and $a_0\in J$ is automatically distinct from the shifts in I. The mark is not a divisor coordinate, so the fibres are as before and Lemma 4.3 applies. Lemma 4.2 with $U_0=J$ gives

$$\mathbb{E}_X(Z^2V_J)=\sum_{j=0}^{k}\frac{\|f_j\|_2^2}{j!}\frac{|J|\,|I|^j}{L^{j+1}}+o(1)\longrightarrow\eta\sum_{j=0}^{k}\alpha_j\|f_j\|_2^2=\eta w.$$

Marked moment in I. For $a\in I$, Lemma 3.1 gives the identity $\vartheta(m+a)Z(m)^2=\vartheta(m+a)Z_a(m)^2$ for all m (both sides vanish when $m+a$ is composite). Thus

$$\mathbb{E}_X(Z^2V_I)=L^{-1}\sum_{a\in I}\mathbb{E}_X\big[\vartheta(m+a)Z_a(m)^2\big],$$

and Z_a involves only subsets $U\subset I\setminus\{a\}$, so the mark $a_0=a$ is distinct from all divisor shifts. The functions $\widetilde{F}_j$ have budget τ and, by Lemma 3.1, signed derivatives f_j+Tf_{j+1} vanishing on coordinate faces for $j\geq 1$. Lemmas 4.2 and 4.3 apply with $U_0=I$ and all $U_i=I$. For $U=U'$ of size j the constant is $\|f_j+Tf_{j+1}\|_2^2$, and the shift tuples $(a,u_1,\dots,u_j)$ are exactly the distinct $(j+1)$-tuples in I. Hence

$$\mathbb{E}_X(Z^2V_I)=\sum_{j=0}^{k}\frac{\|f_j+Tf_{j+1}\|_2^2}{j!}\frac{|I|^{j+1}}{L^{j+1}}+o(1)\longrightarrow\lambda\sum_{j=0}^{k}\alpha_j\|f_j+Tf_{j+1}\|_2^2=v.$$

(For $j=0$ the factor is the scalar $\widetilde{F}_0=f_0+\int_0^\infty f_1$.)

Fourth moment. $\mathbb{E}_X Z^4$ is a sum over quadruples of subsets of I; the budgets sum to $4\tau=1/2<1$, there are finitely many patterns, and by Lemma 4.2 each contributes $O(1)$ (we only need finiteness of $\mathcal{C}_\kappa$, which Proposition 3.2 provides for fibres of any size).

Pair majorant. We have

$$\mathbb{E}_X(Z^2\Pi_J) = \sum_{\substack{b,c\in J\\ b\neq c}} \sum_{j,j'} \sum_{S,T} \mathbb{E}_X\big[D_{F_j}(m;S)D_{F_{j'}}(m;T)\, D_G(m;b)^2 D_G(m;c)^2\big].$$

This is unmarked, with six factors of budget τ, total $6\tau = 3/4 < 1$. The shifts $b \neq c$ lie in J and are distinct from all shifts in I; each of b and c forms a fibre of size exactly two (the two copies of D_G), while the fibres from S, T have size one or two. By Lemma 4.3, only $S = T$ contributes to the main term. For $S = T$, $|S| = j$, the signed derivative of the one-dimensional G is $-G'$, so by (11) and Fubini

$$\mathcal{C}_\kappa = \|f_j\|_2^2\Big(\int_0^\infty G'(y)^2 dy\Big)^2 = c_G^2\|f_j\|_2^2.$$

The shift tuples are $(b, c, a_1, \dots, a_j)$, distinct, with $b, c \in J$ ordered and $a_i \in I$; here $t = j + 2$ and $\mathfrak{n}_\kappa = j!$. Lemma 4.2 with U's equal to $J, J, I, \dots, I$ gives

$$\mathbb{E}_X(Z^2\Pi_J) = c_G^2 \sum_{j=0}^{k} \frac{\|f_j\|_2^2}{j!} \frac{|J|^2|I|^j}{L^{j+2}} + o(1) \longrightarrow c_G^2\eta^2 w.$$

This completes the proof of Proposition 4.1. □

Remark 4.4. The last computation is the off-diagonal part of the proof of the detector bound [6, (7)], where it is shown that the same sum (with J and I of equal length) tends to $\lambda^2 c_G^2 w$. Two features matter for us. First, the constant c_G^2 depends only on G, not on the family $\mathbf{f}$, on k, or on the length of I. Second, the off-diagonal term is of order η^2 in the relative length of J, while the diagonal term in [6, (7)] (and our main term $\mathbb{E}_X(Z^2V_J)$) is of order η. Since, exactly as in the proof of [6, (7)], $Z^2V_J^2 \le q_X Z^2 V_J + q_X^2 Z^2\Pi_J$ pointwise, Proposition 4.1 also recovers the detector bound $\limsup \mathbb{E}_X(Z^2V_J^2) \le (\eta + c_G^2\eta^2)w$ in our configuration.

5 Proof of Proposition 2.1

Fix $\eta > 0$, $R \ge 1$ and $\varepsilon > 0$, and put $\lambda = 2R\eta$. By Proposition 3.5 applied with this λ, there is a single admissible family $\mathbf{f}$ (with some finite k) such that

$$w = w(\mathbf{f}) \ge \tfrac{1}{2} \qquad \text{and} \qquad v(\mathbf{f}) \le \varepsilon w.$$

Freeze this family and define

$$W_X(m) = \frac{Z(m)^2}{w} \ \ge 0 \qquad (X < m \le 2X).$$

Dividing (12)–(14) by w gives $\mathbb{E}_X W_X \to 1$, $\mathbb{E}_X(W_X V_J) \to \eta$, $\mathbb{E}_X(W_X V_I) \to v/w \le \varepsilon$ and $\mathbb{E}_X(W_X\Pi_J) \to c_G^2\eta^2$; and $\mathbb{E}_X W_X^2 = \mathbb{E}_X Z^4/w^2 = O(1)$, with an implied constant depending on the frozen family, hence on η, R, ε. This proves Proposition 2.1. □

The order of the choices is the same as in [O] and is essential: the auxiliary limit producing the family (in which $k \to \infty$) is taken first, and only then $X \to \infty$. No estimate uniform in k is needed. In the proof of Theorem 1.1 the choices are: G (absolute), then $\eta = 1/(2c_G^2)$ (absolute), then R (depending on C_1, C_2), then $\varepsilon = \eta/4$, then the family (depending on $\lambda = 2R\eta$ and ε), then X. The only quantity that must be independent of the family is the coefficient c_G^2 in (5), and it is.

6 Remarks on the limits of the method

The remarks in this section are heuristic and are not used elsewhere.

Remark 6.1 (Where the primes come from). In the argument above, primes are produced only through the *first* moment $\mathbb{E}_X(W_X V_J)$, which is a linear functional of the primes in a block J disjoint from the shift set of the weight. Everything else, namely the suppression in I, the factorial moment of N_J, and the passage from weighted to ordinary counts, enters through upper bounds. The Bombieri–Vinogradov theorem permits an asymptotic formula for weighted averages with a single prime mark (the support condition $\sum \rho_r < 1/2$ in Proposition 3.2); averages with two prime marks are out of reach, as for the twin prime problem. Any pattern that can be forced by *one* prime together with emptiness conditions is accessible in principle; Theorem 1.1 is the natural two-sided instance.

Remark 6.2 (Three consecutive large gaps). Three consecutive gaps $p_n - p_{n-1}$, $p_{n+1} - p_n$, $p_{n+2} - p_{n+1}$ all exceeding $C \log p_n$ require two consecutive primes, each isolated from the outside, at mutual distance exceeding $C \log p_n$. In the present framework this amounts to bounding from below the weighted probability that two disjoint blocks J_1, J_2, separated and flanked by suppressed blocks, each contain a prime. The direct approach needs a lower bound for $\mathbb{E}_X\big(W \mathbf{1}_{\{N_{J_1} \geq 1\}} \mathbf{1}_{\{N_{J_2} \geq 1\}}\big)$, a correlation of two prime events, which at the level of asymptotic formulas is a two-mark problem of twin-prime type, which the lower bounds used here do not reach (we expect parity-type obstructions, but do not claim a proof of this). The indirect approach $\mathbf{1}_{A \cap B} \geq \mathbf{1}_A + \mathbf{1}_B - 1$ would require weighted probabilities summing to more than one. With weights of the present type, which are built from divisors of shifts in I only, one has $\mathbb{E}_X(W \mathbf{1}_{\{N_J \geq 1\}}) \leq \mathbb{E}_X(W N_J) \to |J|/L$ (asymptotically), while the lower bound supplied by the detector estimate, $(\mathbb{E}_X W V_J)^2 / \mathbb{E}_X(W V_J^2) \approx \eta/(1 + c_G^2 \eta)$, never exceeds $1/c_G^2 \leq 1/64$. Lengthening J increases the expected number of primes in J, but the available lower bounds for the *probability* of a prime in J stay below $1/64$, so with the lower bounds used here the two probabilities cannot be shown to sum to more than one. Making these probabilities large would require incorporating the shifts of J_1, J_2 into the weight, as in the Maynard–Tao method [5]; but that method locates several primes among many admissible positions without controlling *which* positions, and we do not see how to combine it with the localization into two separated blocks and the emptiness of the blocks between and around them. We therefore leave three (or more) consecutive gaps larger than $C \log p$, with positive lower density, as an open problem.

Remark 6.3 (A large gap next to a small gap). A large gap adjacent to a *bounded* gap cannot occur with positive density at all: by the upper-bound sieve, for each fixed H the number of $n \leq N$ with $p_{n+1} - p_n \leq H$ is $O_H(N/\log N)$. The natural variant is to ask for a positive proportion of n with $p_n - p_{n-1} < \theta \log p_n$ and $p_{n+1} - p_n > C \log p_n$, for fixed small $\theta > 0$ and large C. This again requires two primes, p_{n-1} and p_n, in prescribed relative positions, together with an empty interval, and the discussion of the previous remark applies: the present method produces one prime per detected configuration. We do not know how to reach this with the techniques used here.

Remark 6.4 (Asymmetric and multi-scale variants). The proof gives slightly more than stated. For fixed integers $R_1, R_2 \geq 1$ one may use R_1 blocks on the left and R_2 on the right; then $\lambda = (R_1 + R_2)\eta$ and the conclusion is that the unique prime of $m + J$ has gaps exceeding $R_1 h$ and $R_2 h$ to its left and right. Taking $R_1 = R_2$ as we did is only for notational convenience, and it loses nothing for Theorem 1.1.

Acknowledgement of method

The method of this paper, including all analytic estimates, is that of the OpenAI preprint [6]. The present contribution consists of the three-block adaptation, the exactly-one-prime detection via

the off-diagonal detector term, the block form of the singular-series average, and the resulting consequences (Theorem 1.1 and Corollaries 1.2 and 1.3).

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